and power cables. We also observed the sturdiness of the CD/DVD drive tray, the quality and feel of the buttons on the cabinet, the mouse and the keyboard, labelled ports at the rear of the system, etc.

c) Upgradability: Here we logged specifications such as the extent to which the processor could be upgraded (the highest clock speed and technology), the maximum amount of RAM that could be added, the number of PCI slots on the motherboard and the number of ports (USB, serial) available in the system.

d) Package contents: Since the documentation and software bundle play an important role in the overall value of a branded system, we logged the inclusion or exclusion of components such as the operating system installer CD or restore disk, the various bundled software, driver CDs, manuals, setup guides, etc.

Performance: To determine the overall performance of the PCs, we used a battery of software ranging from low-level tests for the system's subcomponents, to real world applications for evaluating the performance of a machine in specific usage areas.

SiSoft Sandra 2002 SE was the main benchmarking tool for evaluating the system's component-level performance. It examined the performance of components such as the system processor, the hard disk drive and CD/DVD drive (data transfer rates and access times) and the RAM bandwidth. Virtual Dub was used to evaluate DivX video encoding and Audio Catalyst for MP3 encoding. We logged the time taken by the system to process a sample test file in each of these cases. These tests determined the capability of the computers in handling large amounts of streaming data. On the gaming front, we used *Quake III Arena* version 1.31 and *Evolva Scout Bump Rolling* demo for ascertaining the performance of a computer under OpenGL and Direct3D, respectively. The final real world test was related to image processing. Here, we ran a set of 21 filters in Photoshop 7 and logged the time taken to complete the task. This test provided an insight into the performance of the processor, RAM and hard disk.

We also tested a set of important subsystems in the computer and the peripherals that came with it—RMAA was used to evaluate the frequency response, Total Harmonic Distortion and noise in the soundcard and Displaymate was used for evaluating the sharpness and resolution of the monitor. The speakers were tested for music and movie performance.

A special score for each of the PCs, called the Suitability to Task Index (STTI), was finally computed. This index was calculated using a different set of weightages for each of the tests according to the effect they would have on three prime application areas: productivity, gaming and entertainment, and office and Internet applications. This index provided a clear indication of the performance of each PC in these application areas. For example, the STTI for productivity was calculated by allotting a greater weightage to tests such as the Photoshop filter test than to the gaming tests. Similarly, for gaming and entertainment, a greater importance was allotted to the *Quake III* and the speaker tests.

The Suitability to Task Index is an indication of the PC's performance in each of the three application areas they were tested in. Higher scores indicate the PC being more suited to that particular application area

Value for money: This index was computed by evaluating the system's performance coupled with the features, in relation to the price: the higher this index, the better a product scored.

Warranty and support: Here, we logged specifications such as the type and duration of warranty offered by the manufacturer, the facility of an Annual Maintenance Contract (AMC), presence of a call centre, the average time taken to attend to a system problem and the number of service centres present in the country. We also allotted extra points if the manufacturer offered finance options for purchasing the computer.

How they Fared

Desktop computers used to be chunky devices that were purely utilitarian. But PCs have now evolved to include high-speed processors, faster and larger hard disks, cheaper and faster RAM and supercharged 3D graphics accelerators—all this allows you to use your PC not just for simple computing but also for entertainment and for professional applications. Here's how the PCs in this comparison fared in terms of the features they offer and their suitability to different application areas.

A task as simple as playing an MP3 file requires the processing power of at least a 133 MHz CPU in order to run smoothly. A few years ago, we would have been shocked at a single entertainment application taking up so much processing power. But

Bazaar Buzz

Over 20 lakh PCs are sold in India in a year, a majority of them in the metros. However, due to the general economic slow down, the rate of sale is not as high as expected. Corporate customers, who contribute to a majority of the sales, are resorting to either upgrades or the assembled variety. Government purchases have minimised to the extent that they are practically non-existent. The SoHo and home segment accounts for a major sales chunk. The local dealer market has been slowly regressing due to stiff competition from bigger brands and piracy is rampant with assemblers relentlessly loading whichever operating system and application software on their PCs. However, the recent efforts of organisations such as NASSCOM have delivered very perceptible results in curbing this trend.

MNC brands still command a higher market share than the Indian ones, due to the various bundled applications and their prompt service and quality assurance. This spells advantages, especially for those who place a high premium on quality and service.

The hot favourite MNC brands are Dell, which has excellent upgradability options; Compaq, an eternal favourite with the home segment; and IBM, which is doing well in the corporate world with its PCs specifically tailored for business users.

However, several Indian vendors have developed their own strategies to counter the foreign powers. Unlike last year, where Indian brands were primarily large dealers with a national presence, today after-sales support and service, licensed and free software and peripheral bundles are given primacy by even the local vendors.